

15.1 Ideal Gas Law

Question Paper

Course	CIE A Level Physics
Section	15. Ideal Gases
Topic	15.1 Ideal Gas Law
Difficulty	Medium

Time allowed: 70

Score: /53

Percentage: /100

Question 1a

The ideal gas equation in terms of the number of molecules, N is written as

$$pV = nRT$$

where k is the Boltzmann constant.

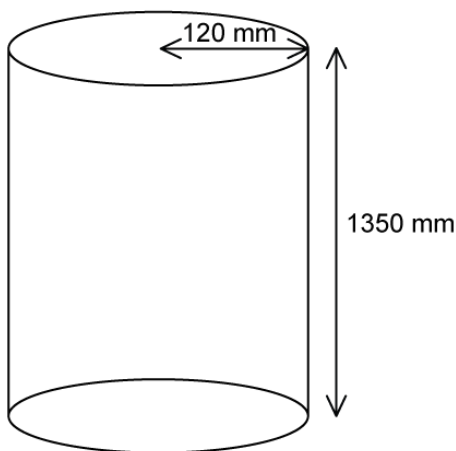
Show that the ideal gas equation can also be written as

$$pV = NkT$$

[2 marks]

Question 1b

A cylinder for an ideal gas has a radius of 120 mm and a height of 1350 mm. The gas is at a temperature of 14 °C and a pressure of 1.7×10^7 Pa.



Show that the amount of gas in the cylinder is 430 mol.

[3 marks]

Question 1c

The gas leaks slowly from the cylinder at an average rate of 8.3×10^{15} atoms s^{-1} so that, after a time of 51 days, the pressure reduces. The temperature remains constant.

Calculate the percentage of the number of moles lost from the cylinder.

[3 marks]**Question 2a**

Calculate the number of molecules in a gas containing 120 g of $^{16}_8\text{O}$.

[2 marks]**Question 2b**

A different ideal gas has an initial volume of $6.00 \times 10^4 \text{ m}^3$ at a pressure of $1.42 \times 10^5 \text{ Pa}$ and a temperature of 8°C

It is heated at a constant volume so that, in its final state, the pressure is $1.70 \times 10^5 \text{ Pa}$ at a temperature of 63°C .

Show that these two states prove it behaves as an ideal gas.

[3 marks]

Question 2c

Atoms of a real gas each have a diameter of 0.12 nm.

Estimate an accurate value for the volume occupied by 0.54 moles of this gas.

volume = m³

[3 marks]

Question 2d

Explain the differences in the volumes of the atoms in part (b) and part (c).

[3 marks]

Question 3a

(i)

State what is meant by an *ideal* gas.

[2]

(ii)

Sketch a graph that represents the definition from part (i).

[2]

[4 marks]

Question 3b

The air in a bicycle tire has a constant volume of $3.5 \times 10^6 \text{ mm}^3$. The pressure of 0.29 mol of the air is $2.0 \times 10^5 \text{ Pa}$. The air may be considered to be an ideal gas.

Calculate the temperature of the air in the tire.

temperature = °C

[3 marks]

Question 3c

The pressure of the tire is increased using a pump. On each stroke of the pump, 0.0045 mol of air is forced into the tire.

Calculate the number of moles of air in the tire when the pressure increases to $2.7 \times 10^5 \text{ Pa}$ and the temperature increases by 10 °C.

[2 marks]

Question 3d

Calculate the number of strokes of the pump required to increase the pressure to $2.7 \times 10^5 \text{ Pa}$ and the temperature by 10 °C.

[2 marks]

Question 4a

State the Pressure law of ideal gases.

[2 marks]

Question 4b

The pressure exerted by an ideal gas of 8.4×10^{20} molecules in a container of volume $1.6 \times 10^{-5} \text{ m}^3$ is $3.2 \times 10^5 \text{ Pa}$.

Calculate the temperature of the gas in the container in $^{\circ}\text{C}$.

[3 marks]

Question 4c

The pressure of the gas is measured at different temperatures whilst the volume of the container and the mass of the gas remain constant.

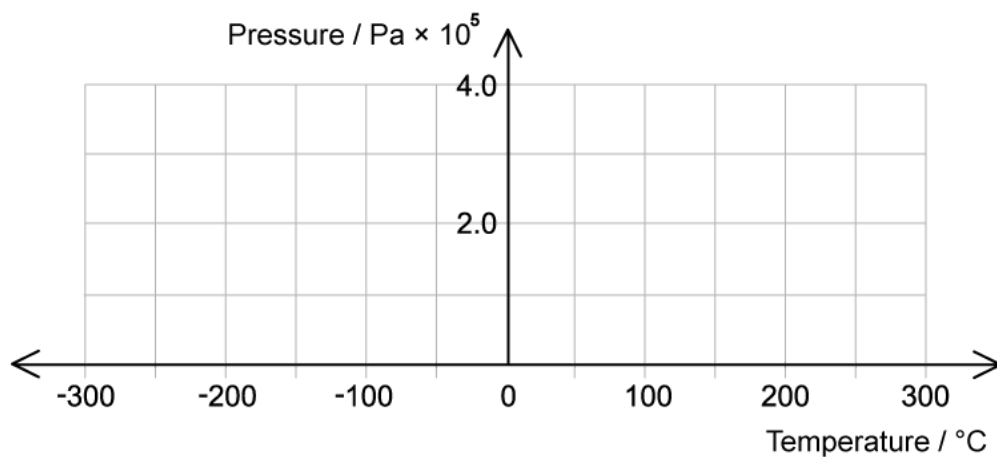


Fig 1.1

Draw a graph on the grid in **Fig 1.1** to show how the pressure varies with the temperature.

[3 marks]

Question 4d

The container described in part (b) has a release valve that allows gas to escape when the pressure exceeds 4.0×10^5 Pa.

Calculate the number of gas molecules that escape when the temperature of the gas is raised to 520°C .

[3 marks]

Question 5a

State two equations or laws which ideal gases obey.

[2 marks]

Question 5b

A car tyre of volume $4.2 \times 10^{-2} \text{ m}^3$ contains air at a pressure of 300 kPa and a temperature of 290 K. The mass of one mole of air is $3.1 \times 10^{-2} \text{ kg}$.

Assuming that the air behaves as an ideal gas, calculate:

(i) The amount of moles of air.

[3]

(ii) The mass of the air.

[1]

(iii) The density of the air.

[1]

[5 marks]

Question 5c

A bicycle tyre with 0.47 moles of air has a volume of $1.90 \times 10^{-3} \text{ m}^3$ when the temperature is 252 K.

Calculate the pressure inside the bicycle tyre.

[2 marks]

Question 5d

After the bicycle has been ridden, the temperature of the air in the tyre is 301 K.

Calculate the new pressure in the tyre assuming the volume is unchanged.

Give your answer to an appropriate number of significant figures.

[3 marks]



Model Answers: Medium

1a

(a) The ideal gas equation can also be written as $pV = NkT$ by:

State the definition of the Boltzmann constant, k :

- $k = \frac{R}{N_A}$

Determine the equation for the number of moles, n :

- $n = \frac{N}{N_A}$

Substitute this into the ideal gas equation:

- $pV = nRT = \left(\frac{N}{N_A}\right)RT$ [1 mark]

Substitute the Boltzmann constant:

- $k = \frac{R}{N_A} \Rightarrow R = kN_A$
- $pV = \left(\frac{N}{N_A}\right) \times kN_A \times T = NkT$ [1 mark]

[Total: 2 marks]

1b

(b) Show that the amount of gas in the cylinder is 430 mol

List the known quantities:

- Radius of the cylinder, $r = 120 \text{ mm} = 120 \times 10^{-3} \text{ m}$
- Height of the cylinder, $h = 1350 \text{ mm} = 1350 \times 10^{-3} \text{ m}$

- Temperature of the gas, $T = 14\text{ }^{\circ}\text{C} = 14 + 273 = 287\text{ K}$
- Pressure, $p = 1.7 \times 10^7\text{ Pa}$

Calculate the volume of the gas, V :

- $V = A \times h = \pi r^2 \times h$
- $V = \pi(120 \times 10^{-3})^2 \times (1350 \times 10^{-3}) = 0.061\text{ m}^3$ [1 mark]

Rearrange the ideal gas equation for the number of moles, n :

- $pV = nRT \Rightarrow n = \frac{pV}{RT}$

Substitute in the values:

- $n = \frac{(1.7 \times 10^7) \times 0.061}{8.31 \times (287)}$ [1 mark]
- $n = 434.8 = 430\text{ mol}$ [1 mark]

[Total: 3 marks]

You will be expected to remember the volume of common 3D shapes such as cubes, cylinders and spheres.

T in the ideal gas equation is **always** in **Kelvin**, do not miss out this conversion if you are given a temperature in $^{\circ}\text{C}$.

1c

(c) The percentage of the number of moles lost from the cylinder is:

- Rate of loss of atoms $= 8.3 \times 10^{15}\text{ atoms s}^{-1}$
- Time, $t = 51\text{ days}$
- Initial number of moles, $n = 434.8\text{ mol}$ (from part b)

Calculate the number of lost atoms:

- $\text{atoms lost} = (8.3 \times 10^{15}) \times (51 \times 24 \times 60 \times 60)$
- $\text{atoms lost} = 3.657 \times 10^{22}\text{ atoms}$ [1 mark]

Calculate the number of lost moles, n :

- $n = \frac{3.657 \times 10^{22}}{6.02 \times 10^{23}} = 0.059 \text{ mol}$ [1 mark]

Calculate the percentage of the number of moles lost:

- $\% \text{ moles lost} = \frac{0.059}{434.8} \times 100\%$
- $\% \text{ moles lost} = 0.01356\% = 0.014\%$ [1 mark]

[Total: 3 marks]

If the rate is in s^{-1} then the time you must multiply this by must be in **s** so the final units are **atoms**. Take care for small unit conversions like this, which are very easy to lose marks from.

2a

(a) The number of molecules in a gas containing 120 g of $^{16}_8\text{O}$ is:

List the known quantities:

- Mass of sample, $m = 120 \text{ g}$
- Molar mass of oxygen-16, $M_r = 16 \text{ g mol}^{-1}$

Calculate the number of moles, n :

- $n = \frac{m}{M_r} = \frac{120}{16}$
- $n = 7.5 \text{ mol}$ [1 mark]

Calculate the number of molecules, N :

- $n = \frac{N}{N_A} \Rightarrow N = nN_A$
- $N = 7.5 \times (6.02 \times 10^{23}) = 4.5 \times 10^{24}$ [1 mark]

[Total: 2 marks]

The molar mass is **always** the **mass number** of the element, in g mol^{-1} . You must be able to read the ^A_ZX notation to correctly read the mass number for any element.

2b

(b) The two states prove it behaves as an ideal gas because:

List the known quantities:

- Initial (and final) volume, $V_1 = 6.00 \times 10^4 \text{ m}^3$
- Initial pressure, $p_1 = 1.42 \times 10^5 \text{ Pa}$
- Initial temperature, $T_1 = 8^\circ\text{C} = 8 + 273 = 281 \text{ K}$
- Final pressure, $p_2 = 0.91 \times 10^5 \text{ Pa}$
- Final temperature, $T_2 = 63^\circ\text{C} = 63 + 273 = 336 \text{ K}$

Determine an equation linking p , V and T :

- $pV \propto T \Rightarrow \frac{pV}{T} = \text{constant}$

Substitute in the initial values:

- $\frac{p_1 V_1}{T_1} = \frac{(1.42 \times 10^5) \times (6.00 \times 10^4)}{281} = 3.03 \times 10^7 \text{ [1 mark]}$

Substitute in the final values:

- $\frac{p_2 V_2}{T_2} = \frac{(1.70 \times 10^5) \times (6.00 \times 10^4)}{336} = 3.03 \times 10^7 \text{ [1 mark]}$

Conclusion:

- This is an ideal gas because $\frac{pV}{T} = \text{constant}$; [1 mark]

[Total: 3 marks]

The temperature must be in **Kelvin** when using the ideal gas equation.

Anytime you want to change a proportionality into an equation:

- Turn the proportionality sign into an = sign and add a **constant** (e.g. k) $pV = kT$

2c

Any time you want to change a proportionality into an equation:

- Turn the proportionality sign into an = sign and add a **constant** (e.g. k) $pV = kT$

2c

(c) An estimate for the volume occupied by 0.54 moles of the gas is:

List the known quantities:

- Diameter of each atom, $d = 0.12 \text{ nm} = 0.12 \times 10^{-9} \text{ m}$
- Number of moles of the gas, $n = 0.54 \text{ mol}$

Calculate the volume of 1 atom:

- *volume of a sphere* $= \frac{4}{3} \pi r^3$
- *volume of 1 atom* $= \frac{4}{3} \pi \left(\frac{0.12 \times 10^{-9}}{2} \right)^3 = 9.05 \times 10^{-31} \text{ m}^3$ [1 mark]

Calculate the number of atoms:

- $n = \frac{N}{N_A} \Rightarrow N = nN_A$
- $N = 0.54 \times (6.02 \times 10^{23}) = 3.25 \times 10^{23} \text{ atoms}$ [1 mark]

Estimate the volume occupied by 0.54 moles of this gas, V :

- $V = \text{volume of 1 atom} \times \text{number of atoms}$
- $V = (9.05 \times 10^{-31}) \times (3.25 \times 10^{23})$
- $V = 2.941 \times 10^{-7} = 3.0 \times 10^{-7} \text{ m}^3$ [1 mark]

[Total: 3 marks]

An atom can be estimated to be a sphere (even if it is not really this shape). In a similar way, it is assumed a wire has a circular cross-sectional area, which will be useful in the topic of resistivity and Hooke's law.

2d

(d) The differences in the volumes in part **(b)** and part **(c)** are:

- The volume in part (b) is for an ideal gas whilst the volume in (c) is as a real / non-ideal gas; [1 mark]
- The volume in part (b) / ideal gas is much larger than the volume of the real gas; [1 mark]
- This is justified because the volume of molecules / atoms is negligible **OR** the volume of the ideal gas is the volume of the container and not the atoms; [1 mark]

[Total: 3 marks]

3a

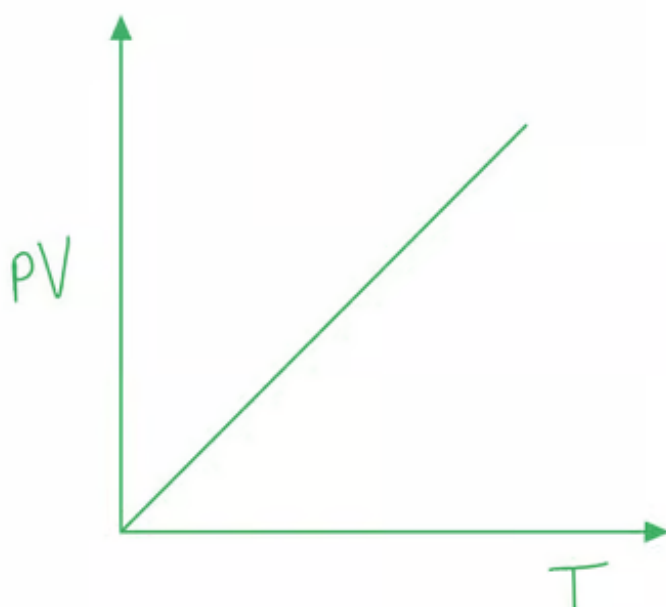
(a)

(i) An ideal gas:

- Obeys the law $pV = \text{constant} \times T$; [1 mark]
- At all values of p , V and T ; [1 mark]

(ii) Sketch a graph that represents the definition from part **(i)**

A graph that would score 2 marks should look like:



- Axes labelled pV and T ; [1 mark]
- Straight diagonal line through the origin; [1 mark]

Determine an equation for the new number of moles, n_2 :

- $pV = nRT \Rightarrow n = \frac{pV}{RT}$ where $\frac{V}{R}$ is a constant

- $n = \text{constant} \times \frac{p}{T}$

- $\frac{n_1 T_1}{p_1} = \frac{n_2 T_2}{p_2}$

- $n_2 = \frac{n_1 p_2 T_1}{p_1 T_2} = n_1 \times \frac{p_2}{p_1} \times \frac{T_1}{T_2}$

Substitute in the values:

- $n_2 = 0.29 \times \left(\frac{2.7 \times 10^5}{2.0 \times 10^5} \right) \times \left(\frac{290}{300} \right)$ [1 mark]

- $n_2 = 0.378 = 0.38 \text{ mol}$ [1 mark]

[Total: 2 marks]

Determining the equation for the final number of moles is quite tricky here. The trick is it to see which variables in the equation are **constant**, and then you can convert this into an equation for the variables **before** (with a subscript '1') and the variables **after** (with a subscript '2'). This comes up many times in A-level physics, especially in the ideal gases topic.

3d

(d) The number of strokes of the pump to increase the pressure to $2.7 \times 10^5 \text{ Pa}$ at a temperature increase of 10°C is:

- Initial number moles, $n_1 = 0.29 \text{ mol}$
- Final number of moles, $n_2 = 0.38 \text{ mol}$ (from part c)
- Number of moles of air into tyre on each stroke = 0.0065 mol

Calculate the number of strokes:

- $n_2 = 0.3 / 8 = 0.38 \text{ mol}$ [1 mark]

[Total: 2 marks]

Determining the equation for the final number of moles is quite tricky here. The trick is it to see which variables in the equation are **constant**, and then you can convert this into an equation for the variables **before** (with a subscript '1') and the variables **after** (with a subscript '2'). This comes up many times in A-level physics, especially in the ideal gases topic.

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- Number of moles of air into tyre on each stroke = 0.0065 mol

Calculate the number of strokes:

- *change in the number of moles* $= n_2 - n_1 = 0.38 - 0.29 = 0.09$ [1 mark]
- *number of strokes* $= \frac{0.09}{0.0065} = 50$ [1 mark]

[Total: 2 marks]

The number of strokes has to be an **integer** so you must round up or down when you calculate a value like this.

4a

The Pressure law of ideal gases is:

- For a constant volume; [1 mark]
- The pressure of a gas is directly proportional to its absolute temperature; [1 mark]

OR

$$\frac{p}{T} = \text{constant}; [1 \text{ mark}]$$

[Total: 2 marks]

4b

List the known quantities:

- Pressure, $p = 3.2 \times 10^5 \text{ Pa}$
- Volume, $V = 1.6 \times 10^{-5} \text{ m}^3$
- Number of molecules, $N = 8.4 \times 10^{20}$

From data booklet:

- The Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
- The Avogadro constant, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
- Molar gas constant, $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$

Method 1: Calculate temperature using the number of molecules, N :

- $pV = NkT$
- $T = \frac{pV}{Nk} = \frac{(3.2 \times 10^5) \times (1.6 \times 10^{-5})}{(8.4 \times 10^{20}) \times (1.38 \times 10^{-23})} [1 \text{ mark}]$
- $T = 441.68 \text{ K} [1 \text{ mark}]$

OR

Method 2: Calculate temperature using number of moles, n :

- $pV = nRT$
- $n = \frac{N}{N_A} = \frac{8.4 \times 10^{20}}{6.02 \times 10^{23}} = 1.40 \times 10^{-3} \text{ mol} [1 \text{ mark}]$
- $T = \frac{pV}{nR} = \frac{(3.2 \times 10^5) \times (1.6 \times 10^{-5})}{(1.4 \times 10^{-3}) \times 8.31} = 440.09 \text{ K} [1 \text{ mark}]$

Convert temperature into $^{\circ}\text{C}$:

- $T = 440.09 - 273.15 = 166.94 = 167^{\circ}\text{C} [1 \text{ mark}]$

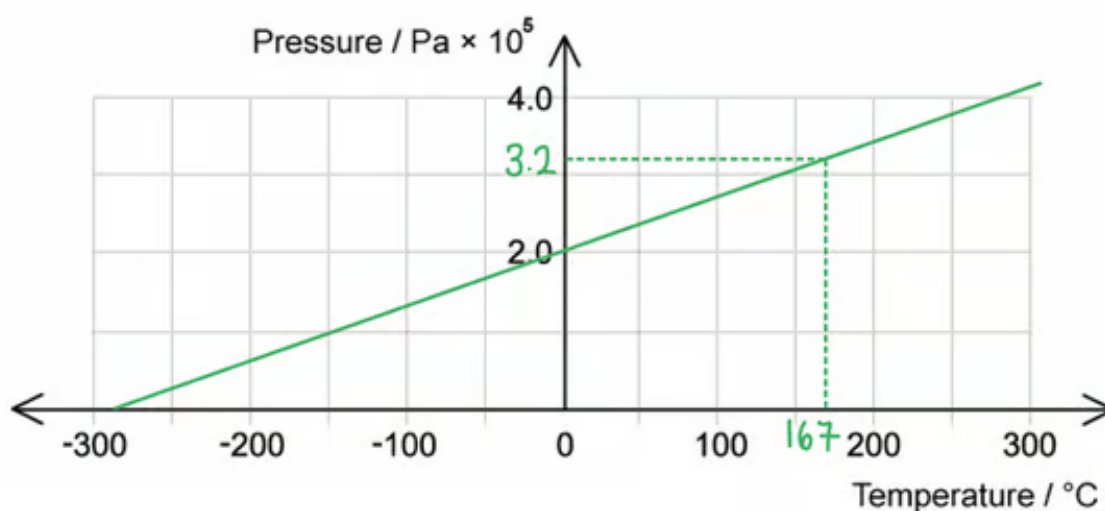
[Total: 3 marks]

4c

List the known quantities:

- Pressure at $167^{\circ}\text{C} = 3.2 \times 10^5 \text{ Pa}$
- Temperature at $3.2 \times 10^5 \text{ Pa} = 167^{\circ}\text{C}$ (from part b)

A graph which would score 3 marks would look like:



- Straight line with positive gradient; [1 mark]
 - This is because the pressure is directly proportional to the temperature at constant volume of an ideal gas (pressure law from part a)
- Through (167, 3.2); [1 mark]
 - This is the temperature 167°C from part (b) and the pressure $3.2 \times 10^5 \text{ Pa}$ given in the equation
- Crosses the temperature axis between -250°C and -300°C ; [1 mark]
 - The temperature scale is in $^{\circ}\text{C}$ therefore absolute zero is at 0 K or -273.15°C

[Total: 3 marks]

Whenever you are asked to sketch or draw a graph, and there are values on the axes, check to see whether there are any specific values you have already calculated that could be relevant.

In this case, we have already calculated a temperature for an equivalent pressure, so

the line for your graph must go through this value.

4d

List the known quantities:

- Pressure, $p = 4.0 \times 10^5 \text{ Pa}$
- Volume, $V = 1.6 \times 10^{-5} \text{ m}^3$
- Initial number of molecules, $N_0 = 8.4 \times 10^{20}$
- Temperature, $T = 520^\circ\text{C} = 520 + 273.15 = 793.15 \text{ K}$

From data booklet:

- The Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$

Number of molecules left in container after valve opens, N :

- $pV = NkT$
- $N = \frac{pV}{kT} = \frac{(4.0 \times 10^5) \times (1.6 \times 10^{-5})}{(1.38 \times 10^{-23}) \times 793.15}$ [1 mark]
- $N = 5.847 \times 10^{20}$ [1 mark]

Number of molecules that will escape:

- Molecules that will escape = initial number of molecules – molecules that have left
- Molecules that will escape = $(8.4 \times 10^{20}) - (5.847 \times 10^{20}) = 2.553 \times 10^{20}$ [1 mark]

[Total: 3 marks]

This question can also be done using the number of moles, n , by rearranging $pV = nRT$. However, remember to find the number of molecules using $N = n$ after!

5a

Two equations that ideal gases obey are:

Any **two** from:

- $pV = nRT$; [1 mark]
- $pV = NkT$; [1 mark]

Number of molecules left in container after valve opens, N :

- $pV = NkT$
- $N = \frac{pV}{kT} = \frac{(4.0 \times 10^5) \times (1.6 \times 10^{-5})}{(1.38 \times 10^{-23}) \times 793.15}$ [1 mark]
- $N = 5.847 \times 10^{20}$ [1 mark]

Number of molecules that will escape:

- Molecules that will escape = initial number of molecules – molecules that have left
- Molecules that will escape = $(8.4 \times 10^{20}) - (5.847 \times 10^{20}) = 2.553 \times 10^{20}$ [1 mark]

[Total: 3 marks]

This question can also be done using the number of moles, n , by rearranging $pV = nRT$. However, remember to find the number of molecules using $N = n$ after!

5a

Two equations that ideal gases obey are:

Any **two** from:

- $PV = nRT$; [1 mark]
- $PV = NkT$; [1 mark]
- Boyle's law / $P \propto \frac{1}{V}$ / $PV = \text{constant}$ at constant temperature; [1 mark]
- Charles's law / $V \propto T$ / $\frac{V}{T} = \text{constant}$ at constant pressure; [1 mark]
- Pressure law / $P \propto T$ / $\frac{P}{T} = \text{constant}$ at constant volume; [1 mark]
- $\frac{PV}{T} = \text{constant}$; [1 mark]

[Total: 2 marks]

5b

List the known quantities:

- Molar mass of air = $3.1 \times 10^{-2} \text{ kg}$
- Volume, $V = 4.2 \times 10^{-2} \text{ m}^3$
- Pressure, $p = 300 \text{ kPa} = 300 \times 10^3 \text{ Pa}$
- Temperature, $T = 290 \text{ K}$

From data booklet:

- Molar gas constant, $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$

(i) The amount of moles of air:

- $pV = nRT$ [1 mark]
- $n = \frac{pV}{RT} = \frac{(300 \times 10^3) \times (4.2 \times 10^{-2})}{8.31 \times 290}$ [1 mark]
- $n = 5.23 = 5.2 \text{ mol}$ [1 mark]

(ii) The mass of the air:

- Mass of the air = molar mass of air $\times$ number of moles of air
- Mass of the air = $(3.1 \times 10^{-2}) \times 5.23 = 0.162 = 0.162 \text{ kg}$ [1 mark]

(iii) The density of the air:

- Density of the air $\rho = \frac{m}{V} = \frac{0.16}{4.2 \times 10^{-2}} = 3.9 \text{ kg m}^{-3}$ [1 mark]

[Total: 5 marks]

5c

List the known quantities:

- Number of moles of air, $n = 0.47 \text{ mol}$
- Volume, $V = 1.90 \times 10^{-3} \text{ m}^3$
- Temperature, $T = 252 \text{ K}$

From data booklet:

- Molar gas constant, $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$

Pressure calculation:

- $pV = nRT$
- $p = \frac{nRT}{V} = \frac{0.47 \times 8.31 \times 252}{1.90 \times 10^{-3}}$ [1 mark]
- $p = 5.18 \times 10^5 \text{ Pa}$ [1 mark]

[Total: 2 marks]

5d

List the known quantities:

- Number of moles of air, $n = 0.47$
- Volume, $V = 1.90 \times 10^{-3} \text{ m}^3$
- Initial temperature, $T_i = 252 \text{ K}$
- Final temperature, $T_f = 301 \text{ K}$
- Initial pressure $p_i = 5.18 \times 10^5 \text{ Pa}$ (from part c)

From data booklet:

- Molar gas constant, $R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$

Determine the correct equation:

- Since the volume is constant, use the Pressure law: $\frac{p}{T} = \text{constant}$

Calculate final pressure, p_f :

- $\frac{p_i}{T_i} = \frac{p_f}{T_f}$
- $p_f = \frac{p_i T_f}{T_i} = \frac{(5.18 \times 10^5) \times 301}{252}$ [1 mark]
- $p_f = 6.19 \times 10^5 \text{ Pa}$ [1 mark] 3 s.f [1 mark]

[Total: 3 marks]